

Fig. 2: Proposed arrangement for compact logic probe

T4 conduct. As a result, the red LED (LED2) glows to indicate logic 0 (low). It can be easily seen from the circuit diagram that a voltage of more than 1.2 volts is required at the probe tip to forward-bias transistors T1 and T3 whose base-emitter junctions appear in series in the circuit. Any voltage less than this at the probe tip will not for-

ward-bias T3, resulting in the red LED (LED1) glowing.

Assemble the circuit on a general-purpose PCB and insert in a small plastic tube, say, a glue stick whose inner mechanism has been removed. For the probe tip, use the front portion of a gel-pen refill. The proposed arrangement for the compact logic probe is shown

in Fig. 2. Before checking the logic of any circuit, connect the black clip to the ground of the circuit and the red clip to the positive terminal of the circuit.

The main feature of the circuit is that when the supply voltage is less than about 4.0V, the red LED (LED2) glows to indicate non-standard TTL voltage supply. ●

STEREO HEADPHONE AMPLIFIER

■ PRADEEP G.

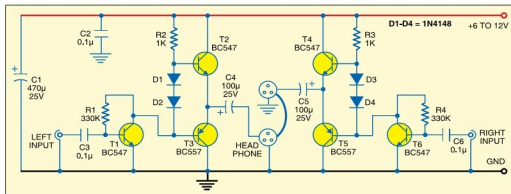
Here is an inexpensive circuit for a stereo amplifier to drive a low-impedance headphone set. The circuit uses a few cheap transistors (BC547 and BC557) and passive components like resistors, diodes and capacitors. It uses one preamplifier stage and npn-pnp push-pull stage to drive headphone.

The preamplifier stage is built around transistors T1 and T6 for left and right inputs, respectively. The amplified left input signal is fed to the push-pull stage built around transistors T2 and T3, which drives the left headphone. Similarly, the amplified right input signal is fed to the push-pull stage built around transistors T4 and T5, which drives the right headphone.

The circuit can output 100-200 mW

at 6V-12V power supply. As current drain is very low, it can be powered from a single 9V PP3 battery also.

Assemble the circuit on a general-purpose PCB and house in a suitable cabinet with the headphone connector. The headphone can be procured from the market.



Carefully solder the components on the PCB to avoid dry soldering. After assembling the circuit, power it with a 9V PP3 battery. The circuit is now ready to use. In case of mains operation, any standard voltage regulator (6V-12V, 100mA) based power supply may be used to power the unit. ●